

Blockchain-Based Environmental Data Integrity for Rural Monitoring Systems

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Abstract

TODO: Write a concise abstract for the manuscript.

1 Introduction

Environmental monitoring systems increasingly rely on distributed sensing infrastructures to collect observations about air quality, water quality, meteorological conditions, and other environmental parameters. Such data support public health assessment, environmental policy, infrastructure planning, agricultural decisions, and local risk management. `TODO:CITATION_NEEDED` In rural monitoring contexts, however, the reliability of environmental data depends not only on sensor accuracy, but also on the integrity of the complete data lifecycle: acquisition, transmission, storage, correction, aggregation, publication, and reuse.

Environmental sensor networks introduce several data integrity challenges. Measurements may be collected by geographically dispersed devices, transmitted through intermittent communication links, buffered locally before synchronization, processed by different organizations, or integrated into public and semi-public datasets. `TODO:CITATION_NEEDED` In rural areas, these challenges can be amplified by limited maintenance access, heterogeneous infrastructure, weaker connectivity, and distributed operational responsibility. Under such conditions, it may be difficult to determine whether a measurement record is original, complete, delayed, corrected, replayed, removed, or modified after collection.

The trustworthiness of environmental data therefore requires more than conventional storage. It requires provenance information that describes where a record originated, which sensor or gateway produced it, when it was collected, which transformations or corrections were applied, and who or what system authorized those changes. `TODO:CITATION_NEEDED` Without such provenance, downstream users may be unable to distinguish between valid corrections, technical failures, missing observations, and unauthorized manipulation. This problem is especially important when environmental data are used for regulatory reporting, infrastructure monitoring, scientific analysis, or public communication. `TODO:CITATION_NEEDED`

Blockchain-inspired audit trails offer one possible approach to this problem. The relevant idea is not cryptocurrency, tokenization, or fully public consensus. Rather, the useful properties are append-only event recording, cryptographic linking of records, timestamped transactions, hash-based verification, and explicit representation of permissions and corrections. `TODO:CITATION_NEEDED` A lightweight blockchain-inspired architecture can store raw environmental measurements off-chain while preserving hashes, audit events, provenance metadata, and verification evidence in an integrity layer. Such a design may support later detection of unauthorized modification without requiring every environmental dataset to be stored directly on-chain.

This direction is motivated by data integrity practices from regulated environments, including clinical research, laboratory systems, and other domains where records must remain attributable, legible, contemporaneous, original, accurate, complete, consistent, enduring, and available. `TODO:CITATION_NEEDED` In these environments, audit trails, controlled corrections, source data verification, role-based authorization, and traceable data custody are treated as core components of trustworthy data management. Although environmental monitoring systems are governed by different technical and legal requirements, the underlying integrity problem is related: collected observations must remain verifiable across organizational, technical, and temporal boundaries.

The research gap addressed in this work is the lack of a clearly defined integrity-oriented framework that adapts regulated audit trail and data provenance concepts to rural environmental monitoring systems. Existing work on environmental sensor networks often focuses on sensing, communication, calibration, data quality, or analytics, while blockchain-oriented environmental systems often emphasize decentralization or transparency. `TODO:CITATION_NEEDED` Less attention has been given to environmental data integrity as a lifecycle problem involving source record identity, timestamped provenance, controlled correction, hash verification, permission changes, audit review, and data custody from collection to publication. `TODO:CITATION_NEEDED`

The primary research question is:

How can regulated data integrity and audit trail concepts be adapted into a blockchain-inspired architecture for preserving and verifying environmental sensor data integrity in rural monitoring systems?

To answer this question, the study pursues the following objectives:

1. Define an environmental data integrity model suitable for rural sensor-based monitoring systems.
2. Identify the environmental data lifecycle events that should be represented in an audit trail.
3. Propose a lightweight blockchain-inspired architecture that separates raw environmental measurements from hashes, audit metadata, permission events, and verification records.
4. Specify verification workflows for detecting unauthorized modification, deletion, replay, broken provenance, and invalid correction events.
5. Design an executable proof-of-concept evaluation using public environmental datasets and controlled tampering scenarios. `TODO:CITATION_NEEDED`

The expected contribution of this paper is a transferable integrity architecture for rural environmental monitoring, grounded in audit trail principles from regulated data environments and implemented through lightweight hash-based verification. The contribution is intentionally scoped: it does not claim that blockchain technology improves environmental outcomes directly, nor does it propose a production blockchain network. Instead, it examines how blockchain-inspired mechanisms can support provenance, auditability, and verification for environmental records whose trustworthiness may otherwise be difficult to establish.

2 Related Work

Summarize relevant literature. Use `TODO:CITATION_NEEDED` for unsupported claims.

3 Methodology

Describe data, methods, assumptions, and evaluation design.

4 System Architecture

Describe the system, workflow, components, and implementation boundaries.

5 Discussion

Discuss implications, limitations, and threats to validity.

6 Conclusions

Summarize findings and outline future work.

References